

Shubo Xu (mini Version)

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Publications

– Publications:

- ◇ X. Li*, **S. Xu***, Z. Zhang, Z. Cai, Y. Chen, and G. Zhai, "DPC-VQA: Decoupling Quality Perception and Residual Calibration for Video Quality Assessment," [arXiv:2604.12813](#), 2026. (**ACMMM'26**)
- ◇ X. Li, Z. Zhang, Z. Xu, **S. Xu**, X. Min, Y. Chen, and G. Zhai, "Decoupling Perception and Calibration: Label-Efficient Image Quality Assessment Framework," [arXiv:2601.20689](#), 2026. (**UR, NeurIPS'26**)
- ◇ **S. Xu**, M. Zhang, W. Song, et al, "A Systematic Review and Analysis of Deep Learning-based Underwater Object Detection," *Neurocomputing*, vol. 527, pp. 204–232, 2023.
- ◇ M. Zhang, **S. Xu**, W. Song, et al, "Lightweight Underwater Object Detection Based on YOLO v4 and Multi-Scale Attentional Feature Fusion," *Remote Sensing*, vol. 13, no. 22, 2021.

– Challenges:

- ◇ H. Zhu, et al., "VQualA 2025 Challenge on Visual Quality Comparison for Large Multimodal Models: Methods and Results," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 3352–3362. [arXiv:2509.09190](#) (**Top 2, VQualA Challenge, ICCV'25**)

Work Experiences

ByteDance Inc.



Aug. 2026 – Present

Senior CV & MLLM Algorithm Engineer

Shanghai, China

Contributed to CV, MLLM, and Agentic applications across TikTok and Doubao ecosystems. Built an MLLM-driven spatial perception system using in-cabin & dashcam imagery for road scene understanding & spatial generation, leveraging VLM post-training, BEV/3D spatial modeling, open-set detection, and OCR/SAM foundation models to power AI Agents.

Baidu Inc.



Jan. 2025 – Jun. 2026

Senior CV & Multimodal Algorithm Engineer

Beijing, China

Contributed to the R&D of AI multimodal search and educational AI systems in AI assistant, with work spanning open-world image understanding, Vision-Grounded RAG, VLM post-training and autonomous vision-language agents.

- **Topic 1: Intent-Aware Fine-Grained Vision-Grounded RAG in Multimodal Dialogues**
- **Topic 2: Feedback-Driven Autonomous Reasoning and Tool Scheduling for Education Agents**
- **Topic 3: End-to-End Joint Object Detection and Instance Priority Recommendation for AI Search**

Hello Ltd.



Jun. 2023 – Dec. 2024

CV Algorithm Engineer

Shanghai, China

Conducted R&D of advanced computer vision algorithms on cloud, mobile and edge platforms, supporting core computer vision techniques for smart city, transportation and risk control applications.

- **Topic 4: Edge-Cloud Collaborative Multi-State Visual Perception for Micro-Mobility Intelligent Transportation Platforms**

Research Interest

★ Human-Centric AI for Future Wellbeing, spanning CV, MLLMs, Agents and AI4Science (e.g., Medical Agents, Emotional & Psychological Support).

Skills

- Programming: Python, C++, C, Objective-C, Java, etc.
- Software Development: Android, iOS, Web, etc.
- DL Technologies: CV, Post-training of VLMs, Agent, Edge-AI, etc.